INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 4

1. Experiment Title:

Combining Conditions

2. Aim:

Combine conditions using AND, filter a range using BETWEEN, and check a list of values using IN.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

AND returns rows only when both conditions are true. BETWEEN checks whether a value lies within an inclusive range. IN checks whether a value matches any value in a supplied list.

5. Algorithm / Procedure:

1. Create the player table and insert the four supplied players.
2. Use AND to find Eagles players with more than 35 runs.
3. Use BETWEEN to find players with runs from 40 through 80.
4. Use IN to find players in the Tigers or Lions teams.

6. Program / Queries / Commands:

```
CREATE TABLE player (
    name VARCHAR(30),
    team VARCHAR(20),
    runs INT
);

INSERT INTO player (name, team, runs)
VALUES
    ('Asha', 'Eagles', 45),
    ('Meera', 'Tigers', 72),
    ('Arjun', 'Eagles', 30),
    ('Kiran', 'Lions', 88);

-- Task 1
SELECT name, runs FROM player
WHERE team = 'Eagles' AND runs > 35;

-- Task 2
SELECT name, runs FROM player
WHERE runs BETWEEN 40 AND 80;

-- Task 3
SELECT name, team FROM player
WHERE team IN ('Tigers', 'Lions');
```

7. Input:

('Asha','Eagles',45), ('Meera','Tigers',72), ('Arjun','Eagles',30), ('Kiran','Lions',88)

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: name	runs
Asha	45
Task 2: name	runs
Asha	45
Meera	72
Task 3: name	team
Meera	Tigers
Kiran	Lions

9. Result:

The SQL statements for Experiment 4 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is the function of AND?
2. Is BETWEEN inclusive of its boundary values?
3. What is the use of IN?
4. Can more than two conditions be combined in WHERE?